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CLIFT: Conformal Self-Verification for Web Agent Training and Test-Time Scaling

A Conformally Certified URL-Aware Verifier Replaces Costly Frontier-LM Supervision with Dense Per-Step Signals and Enables Reliable Test-Time Scaling Without External Oracles

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Web agent training faces a fundamental credit-assignment problem: binary task-success rewards are sparse, while frontier-LM judges providing denser step-level feedback are expensive and unavailable at deployment. CLIFT solves both at once with a conformally calibrated URL-aware verifier that supplies non-negative per-step evidence during training and drives reliable test-time scaling without querying any external oracle.

AT A GLANCE

Problem: Sparse binary task rewards leave most intermediate actions unscored; frontier-LM judges are too expensive for every training step.

Why it matters: Dense, calibrated per-step signals accelerate web agent training and enable reliable test-time compute scaling.

Method: Natural-language verification questions grounded in browser state, calibrated by conformal prediction; same verifier reused at inference for Conformal Test-time Scaling.

Result: Higher task success on WebArena and VisualWebArena; 8–12 pp gain from test-time scaling with $N=8$ trajectories, no oracle needed.

Evidence: WebArena and VisualWebArena benchmarks; ablations on calibration, question quality, and verifier model size.

The Big Picture

Reinforcement learning from task-completion rewards is the dominant paradigm for training web agents, but it suffers a fundamental supervision bottleneck: a twenty-step trajectory that ultimately succeeds returns the same sparse positive signal regardless of which intermediate ac-

tions were actually useful. Frontier language model judges can provide step-level feedback, but querying GPT-4o at every step during training is cost-prohibitive and creates deployment-time dependencies on proprietary APIs.

CLIFT (Conformal Learning for Instruction Following at Test time) addresses both problems simultaneously. During training, it replaces frontier-LM supervision with a small verifier whose outputs are calibrated by conformal prediction, guaranteeing non-negative evidence in expectation. At inference, the same frozen verifier drives Conformal Test-time Scaling (CTS), selecting among candidate trajectories without any external oracle.

Why Existing Approaches Fall Short

Binary rewards are sparse. An agent that completes a task in twenty steps receives no information about which individual steps contributed. This makes gradient estimation difficult and exploration slow.

LM judges are expensive and API-dependent. Querying a frontier model at every step dramatically increases training cost and creates an availability dependency that is absent at deployment.

Heuristic verifiers lack calibration. Rule-based verifiers checking URL patterns or DOM state can fire incorrectly; without calibration their signals may actively mislead training.

Test-time scaling is uncertified. Best-of- N and chain-of-thought approaches for agents offer no statistical guarantee that more compute improves reliability.

Core Mechanism

CLIFT introduces two complementary components:

Conformal Self-Verification (training): Natural-language verification questions (e.g., “Is the agent on the correct URL?”, “Has the required form field been filled?”) are generated from task descriptions and grounded in the current browser state (URL and DOM excerpt). A small verifier answers these questions and assigns per-step scores. Conformal prediction calibrates the verifier’s confidence so it fires only when it can certify its answer at a pre-specified error rate—yielding a conformally certified, URL-aware verifier with a formal coverage guarantee.

Conformal Test-time Scaling (CTS, inference): The same frozen verifier bank scores all candidate trajectories sampled by the agent. CTS selects the trajectory with the highest cumulative conformal evidence, enabling reliable test-time scaling with no frontier-model dependency.

Technical Formulation

Let $s_t = (u_t, d_t)$ be the browser state at step t (URL and DOM excerpt), a_t the agent’s action, and $Q = \{q_1, \dots, q_K\}$ a bank of verification questions derived from the task description.

The verifier model v_ϕ scores each question-state pair:

$$p_{k,t} = v_\phi(q_k \mid s_t, a_t) \in [0, 1]$$

Conformal calibration with coverage $1 - \alpha$ produces a threshold τ such that:

$$\Pr[\text{verifier incorrect}] \leq \alpha$$

The per-step evidence signal used during training is:

$$r_t = \sum_{k=1}^K \mathbf{1}[p_{k,t} \geq \tau] \cdot w_k$$

where w_k are task-specific weights. This signal is always non-negative (evidence of progress, never a penalty), so it supplements rather than replaces the sparse task-completion reward.

For CTS at inference, given N sampled trajectories $\{\tau^{(i)}\}_{i=1}^N$, the selected trajectory is:

$$\hat{\tau} = \arg \max_i \sum_t r_t^{(i)}$$

Learning or Inference Procedure

Training:

1. Generate a bank of verification questions per task type.
2. Calibrate the verifier on a held-out set to determine τ at desired coverage $1 - \alpha$.
3. Run RL with task-completion reward augmented by conformal per-step evidence r_t .

Test-time scaling:

1. Sample N trajectories from the policy.
2. Score each step with the frozen conformal verifier.
3. Return the trajectory with the highest cumulative evidence.

What the Guarantee Says

By construction of conformal prediction, the verifier provides a marginal coverage guarantee: the probability that the verifier incorrectly certifies an incorrect action as correct is at most α (typically $\alpha = 0.1$). This ensures per-step signals are non-negative in expectation, preserving policy improvement under the RL update. The CTS selection guarantee is weaker, but conformal calibration ensures the selected trajectory is at least as likely to be correct as random selection would predict.

Experimental Findings

- **WebArena and VisualWebArena:** CLIFT-augmented RL achieves significantly higher task success than RL with binary reward alone.
- **Evidence density:** The conformal verifier fires at roughly 40–60% of steps, providing substantially denser credit assignment than binary rewards.
- **CTS efficiency:** $N=8$ trajectories with CTS selection improves success rate by **8–12 percentage points** over single-trajectory evaluation, with no frontier-model query.
- **Calibration quality:** The conformally calibrated verifier achieves coverage close to the nominal $1 - \alpha$ across all task types.

Ablations and Interpretation

Calibration vs. uncalibrated verifier: Using the raw verifier without conformal calibration produces noisy per-step signals that can harm training; calibration is critical to the method’s benefit.

Question quality: Generic verification questions degrade performance; the quality of the task-specific question bank governs verifier coverage.

Number of CTS samples: Performance improves with more trajectories up to $N=16$; beyond that, gains are marginal, suggesting the verifier reaches its discrimination ceiling.

Verifier model size: A 1B-parameter verifier delivers comparable quality to a 7B verifier for structured URL-aware verification, indicating that small specialized models are sufficient.

Reference

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